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Tensor-based quantum phase difference estimation for large-scale demonstration

Shu Kanno[a,b,1], Kenji Sugisaki[b,c,d,e], Hajime Nakamura[b], Hiroshi Yamauchi[b,f], Rei Sakuma[b,g], Takao Kobayashi[a], Qi Gao[a,b], and Naoki Yamamoto[b,e,h]

Affiliations are included on p. 7.

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We develop an energy calculation algorithm leveraging quantum phase difference estimation (QPDE) scheme and a tensor-network-based unitary compression method in the preparation of superposition states and time-evolution gates. Alongside its efficient implementation, this algorithm reduces depolarization noise affections exponentially. We demonstrated energy gap calculations for one-dimensional Hubbard models on IBM superconducting devices using circuits up to 32-system (plus one-ancilla) qubits, a five-fold increase over previous Quantum phase estimation (QPE) demonstrations, at the 7242 controlled-Z gate level of standard transpilation, utilizing a Q-CTRL error suppression module. Additionally, we propose a technique toward molecular executions using spatial orbital localization and index sorting, verified linear polyene simulations up to 21 qubits. Since QPDE can handle the same objectives as QPE, our algorithm represents a leap forward in quantum computing on real devices.

quantum computers | quantum chemistry | quantum phase estimation | tensor networks

Significance

Quantum phase estimation (QPE) is a foundational algorithm for quantum chemistry, cryptanalysis, and solving linear equations, due to the potential of exponential acceleration for classical algorithms. Despite its importance, QPE has been limited to a few qubits on real devices due to high gate costs, reinforcing the belief that large-scale QPE requires error-corrected quantum computers. We challenge this by introducing a scalable, noise-robust algorithm tailored for quantum chemistry. The demonstration is given in (quasi) one-dimensional (1D) systems, specifically 1D Hubbard models and molecular systems up to 33 qubits, and the broader applicability of the proposed techniques beyond 1D remains to be demonstrated. These results allow QPE executions of a scale near the classical limit for full configuration interactions.

Abilities to calculate physical properties of materials with high accuracy are crucial for accelerating novel material discovery. The physical properties are mainly governed by the behavior of electrons in the material, while the number of possible electronic state configurations scales exponentially with system size. Approximation methods such as the Hartree–Fock method and density functional theory are commonly used to cope with this exponential increase in classical computers. However, these methods often fail when electronic correlations are strong, leading to computational inaccuracies (1–3). Quantum computers are expected to overcome the accuracy limitations of classical computers because complex quantum states intractable by classical means can be efficiently represented using quantum superposition.

One of the most important quantum algorithms is quantum phase estimation (QPE) (4, 5). QPE aims to find the eigenvalue corresponding to a given eigenstate of a system, representing the energy in quantum chemistry. In the conventional QPE (5), even if the approximate wave function $|\varphi\rangle$ is used as an initial wave function, the full configuration interaction (FCI) energy can be obtained with probability proportional to the squared overlap with the eigenfunction of the target Hamiltonian H. Thus, QPE can potentially compute FCI energies exponentially faster than classical methods, given appropriately prepared $|\varphi\rangle$.

We focus on Bayesian QPE (6), which utilizes Bayesian inference to iteratively narrow the CI of the target phase value. The schematic is shown in Fig. 1*A*; the probability of obtaining 0 in the measurement on the ancilla qubit constitutes a cosine curve for an energy parameter ϵ, the peak of which approximates the target phase. Controlled operations of time evolution gate U_{Trot} often require significant gate overhead in Bayesian (and conventional) QPE. For instance, a recent experiment for a two-qubit hydrogen molecule simulation on a trapped-ion device used 920 two-qubit gates (7). Moreover, most of the QPE studies performed one- or two-qubit systems, and even using classical preprocess with exponential cost, a six-qubit system was the maximum (8–14). The overhead is particularly challenging in nearest-neighbor architectures like superconducting devices, which require qubit swaps for long-range operations. Hence, the quest for gate-efficient implementations is essential for large-scale QPE-type algorithms.

In this work, we propose a gate-efficient, QPE-type energy estimation algorithm. Our algorithm is based on the quantum phase difference estimation (QPDE) (15), a variant of QPE. QPDE can also compute FCI energy (16) but primarily targets the energy gap between states (e.g., ground and excited states) (15, 17, 18). See *SI Appendix*, section S1 for

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[1]To whom correspondence may be addressed. Email: shu.kanno@quantum.keio.ac.jp.

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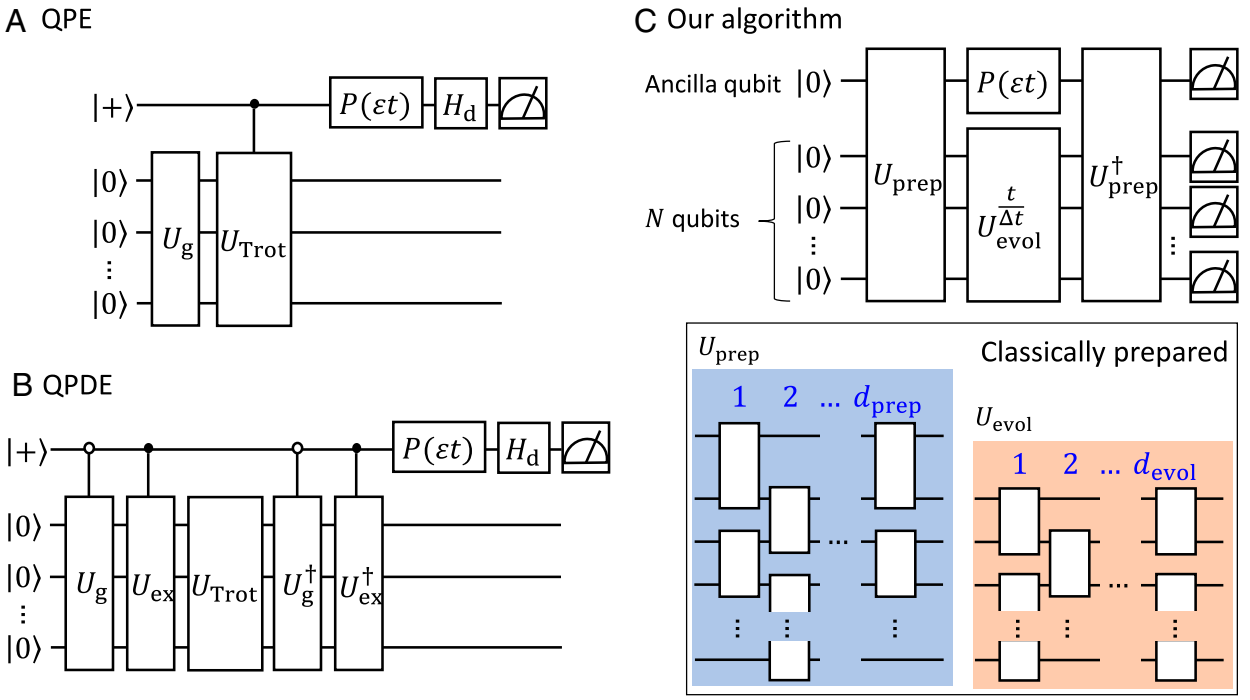


Fig. 1. Circuits for the Bayesian QPE-type algorithms and our algorithm. In all the figures, the topmost qubit is an ancilla qubit and the others are system qubits. $P(x)$ is a phase gate, and H_{d} is a Hadamard gate. U_{Trot} is a gate for the time evolution. U_{g} and U_{ex} are gates for approximate ground and excited state preparations, respectively. U_{prep} is an MPO for approximately preparing the superposition of ground and excited states, and U_{evol} is an MPO for approximating the time evolution operator. (*A*) Quantum phase estimation. (*B*) Quantum phase difference estimation. (*C*) Our algorithm. Note that while we can classically prepare the compressed operator of a single time step U_{evol}, it becomes intractable to classically simulate total time duration $U_{\mathrm{evol}}^{t/\Delta t}$.

details. The important feature of QPDE is an avoidance of costly controlled-U_{Trot} operations. Instead, as shown in Fig. 1*B*, it applies controlled operations to U_{g} and U_{ex} for preparing superpositions of ground and excited states, where U_{g} and U_{ex} are gates for approximate ground and excited state preparations, respectively. Our algorithm is a gate-efficient realization of QPDE; specifically, we classically prepare compressed versions of U_{Trot} as well as the controlled-U_{g} and U_{ex} as tensor networks, particularly matrix product operators (MPOs) (19–28). These classically prepared operators are realized on a quantum circuit shown in Fig. 1*C*. The proposed algorithm offers four main advantages. First, the resulting MPO-based circuits are constructed with nearest-neighbor gates in a brick-wall layout, facilitating parallelization without swaps. Second, the MPO can be efficiently prepared on classical computers and implemented on a quantum circuit, meaning that our algorithm does not suffer from a large amount of quantum-classical communication as in the variational quantum eigensolver (29). Third, because the initial superposition state is efficiently prepared as a matrix product state (MPS), the corresponding state preparation gate can be well approximated by an MPO. Finally, depolarizing noise effects are exponentially suppressed with respect to the number of qubits. The theoretical details can be seen in *SI Appendix*, section S2 for the first three points and *SI Appendix*, section S5 for the final point.

We apply our algorithm to calculate the energy gap for the one-dimensional Hubbard model, utilizing a quantum circuit with up to 32 qubits (plus one ancilla) on an International Business Machines Corporation (IBM) superconducting device, aided by a Q-CTRL error suppression module, where the overview of this module is described on *SI Appendix*, section S1. Notably, this scale is five times larger than previous QPE studies (14). In addition, we investigate linear polyenes of 1,3,5-hexatriene, 1,3,5,7-octatetraene, and 1,3,5,7,9 decapentaene (hereafter referred to as hexatriene, octatetraene, and decapentaene, respectively) as model systems of linear π-conjugated polyenes. In linear polyenes, the character of the lowest excited singlet state is known to change depending on the length of π-conjugation (30), and the chemistry of excited states is of significant importance. In the calculation of hexatriene, a one-dimensional 12-qubit Hamiltonian was constructed by using developed orbital localization and index reordering techniques. We also execute it on octatetraene and decapentaene of 16- and 20-qubit Hamiltonian, respectively, in *SI Appendix*, section S5. Finally, recall that the MPO-based time-evolution gate of short time Δt can be classically implemented, while its concatenation for long-time simulation cannot. Thus, the quantum advantage of this algorithm lies in the time evolution circuit. See *SI Appendix*, section S4 for the numerical verification of exponential growth for bond dimensions in an MPO.

Tensor-Based Phase Difference Estimation Algorithm

The entire estimation algorithm is summarized to *SI Appendix*, Algorithm 1, and here we describe the procedure sketch. For each iteration, the multiple circuits with different parameters ϵ determined from the prior distribution are executed, where the prior and posterior distributions are assumed to be Gaussian, and the total time t is determined from the variance of the prior distribution. In advance to executing the quantum algorithm with the circuit Fig. 1*C*, we compute MPOs corresponding to U_{prep} and U_{evol} by classical means, followed by preparing their gate realization.

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The quantum algorithm is executed as follows; we prepare the superposition state using U_{prep} gate, evolve the state by operating U_{evol} by $t/\Delta t$ times, operate the phase gate with εt, uncompute the state using $U_{\mathrm{prep}}^{\dagger}$ gate, and measure all qubits to obtain the probability of getting all 0. The number of depths of U_{prep} and U_{evol} are d_{prep} and d_{evol}, respectively. We repeat this procedure with different ε to obtain the likelihood function which can be further approximated by Gaussian distribution. The posterior distribution is calculated to update the CI of ε, and the posterior distribution is used as the prior distribution for the next iteration.

We can also calculate the FCI energy by simplifying the procedure. That is, the value of gap with a negative sign corresponds to the FCI energy when $|\psi_{\mathrm{ex}}\rangle$ is a vacuum state (16), and see *SI Appendix*, section S6 for the demonstration of the FCI energy calculation. Note that by extending the tensor structure from an MPO to a complicated tensor such as a tree tensor, our algorithm would be useful even for all-to-all connected devices such as trapped-ion and neutral atom devices since the long-distance transport of ions (or atoms) is costly in practice (31).

Benchmark on the Unitary Compression Method

We provide a brief comparison of the proposed unitary compression method with conventional methods for state preparation and time evolution circuits. We consider the following one-dimensional Hubbard model:

$$H = -T\sum_{q=1}^{n_s-1}\sum_{\sigma\in\{\uparrow,\downarrow\}}\left(a_{q+1\sigma}^{\dagger}a_{q\sigma} + a_{q\sigma}^{\dagger}a_{q+1\sigma}\right) + U\sum_{q=1}^{n_s} n_{q\uparrow}n_{q\downarrow} - \frac{U}{2}\sum_{q=1}^{n_s}(n_{q\uparrow}+n_{q\downarrow}), \quad [1]$$

where q (σ) is the orbital (spin) index, $T = 1$ is the hopping energy so that energy is unitless, U is the on-site Coulomb energy, $a_{q\sigma}^{\dagger}$ ($a_{q\sigma}$) is the creation (annihilation) operator, and $n_{q\sigma}$ is the number operator $n_{q\sigma} = a_{q\sigma}^{\dagger}a_{q\sigma}$. In this study, we choose $U = 10$, which corresponds to the strong correlation regime, and we consider $n_s = 4$, i.e., eight-system qubit model in this section, and the fermion–qubit mapping is a Jordan–Wigner transformation with an up–down qubit sequence $1\uparrow, 1\downarrow, 2\uparrow, \ldots, n_s\downarrow$.

The metric to evaluate the U_{prep} approximation for the state preparation circuit is $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle)$ represented by

$$f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle) = \langle \mathrm{MPS}|\ U_{\mathrm{prep}}\ |0\rangle^{\otimes N+1}, \quad [2]$$

where N is the number of system qubits and $N = 2n_s$ in the Hubbard model. Because we consider only the real wave function in this study, the metric is real. $|\mathrm{MPS}\rangle$ is an MPS calculated for the ground and first-excited states using the density matrix renormalization group (DMRG) (32, 33). Because we confirmed that the DMRG result was almost the same as the exact solution in the current eight-qubit model, $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle)$ is close to the square root of fidelity $f(U_{\mathrm{prep}};\ |\psi_{\mathrm{target}}\rangle)$, where $|\psi_{\mathrm{target}}\rangle = \frac{1}{\sqrt{2}}(|0\rangle\,|\psi_{\mathrm{g}}\rangle + |1\rangle\,|\psi_{\mathrm{ex}}\rangle)$, and $|\psi_{\mathrm{g}}\rangle$ and $|\psi_{\mathrm{ex}}\rangle$ are approximate ground and excited states obtained using the DMRG, respectively. The metric $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle)$ takes the maximum value 1 when $U_{\mathrm{prep}}\,|0\rangle^{\otimes N+1} = |\mathrm{MPS}\rangle$. The coefficient of the Hartree–Fock configuration in the exact ground state is 0.66. On the other hand, the $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle)$ for this model was 0.99 at $d_{\mathrm{prep}} = 6$, which indicates that the U_{prep} can prepare states with high accuracy.

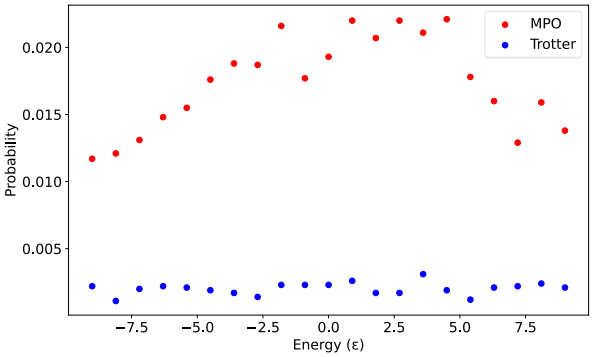


Fig. 2. Comparison of the time evolution circuits (nine qubits) of the first-order Trotter approximation and that of the MPO. The Eagle device *ibm_osaka* was adopted. The probability of all 0 measurements for each of the phase ε is plotted.

The error metric of the U_{evol} approximation in the time evolution circuit is $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ as in ref. 26:

$$\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}}) = \sqrt{2 - \mathrm{Re}\,\mathrm{Tr}\left[U_{\mathrm{ref}}^{\dagger}U_{\mathrm{evol}}\right]^{\frac{1}{N}}}, \quad [3]$$

where U_{ref} is a reference MPO prepared by converting the time evolution operator to an MPO (see *SI Appendix*, section S2 for details). $\frac{1}{N}$ is the normalization factor. From the related reason in the state preparation of $|\mathrm{MPS}\rangle \approx |\psi_{\mathrm{target}}\rangle$, $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ is close to $\delta(U_{\mathrm{evol}};\ e^{-iH\Delta t})$ in the current model. We choose $\Delta t = 0.1$ throughout this study. $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ takes the minimum value 0 when $U_{\mathrm{evol}} = U_{\mathrm{ref}}$. $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ at $d_{\mathrm{evol}} = 5$ was 4.3×10^{-3}, and $\delta(U_{\mathrm{evol}};\ e^{-iH\Delta t})$ of the first- and second-order Trotter approximations were 2.2×10^{-2} and 1.6×10^{-3}, respectively. The values suggest that the accuracy of prepared U_{evol} is between first- and second-order Trotter approximations, which corresponds to the tendency in the previous study (26).

To confirm the circuit execution efficiency, we compared the results of the measured probability distributions in the circuit of Fig. 1*C* run on a real device with U_{evol} as is and replaced by the first-order Trotter approximation. The variance of the prior distribution is 9.0 which corresponds to $t = 0.2$ (*SI Appendix*, section S1), the device is Eagle *ibm_osaka*, $d_{\mathrm{prep}} = 6$, and the error suppression module appeared in the subsequent sections is not used here. The number of shots is 10,000 in the real device execution of this study. The number of two-qubit gates in the circuit after transpilation in Qiskit (34) was 216 for U_{prep} and 234 for the first-order Trotter approximation. Here, we consider the two-qubit gates as the ECR gate and the controlled-Z gate in the Eagle and Heron devices, respectively. The results are shown in Fig. 2. The plots by U_{evol} (red) show peaks, while the Trotter results (blue) show a uniform distribution. These results indicate that the unitary compression has a high execution efficiency for real devices.

Demonstration of the Algorithm

We show the demonstration for the Hubbard model in Eq. **1**. Fig. 3*A* shows the result of a noiseless simulator for our algorithm with $n_s = 4$, i.e., a nine-qubit circuit including one ancilla qubit, where $d_{\mathrm{prep}} = 6$ and $d_{\mathrm{evol}} = 5$ which are the same setting in the

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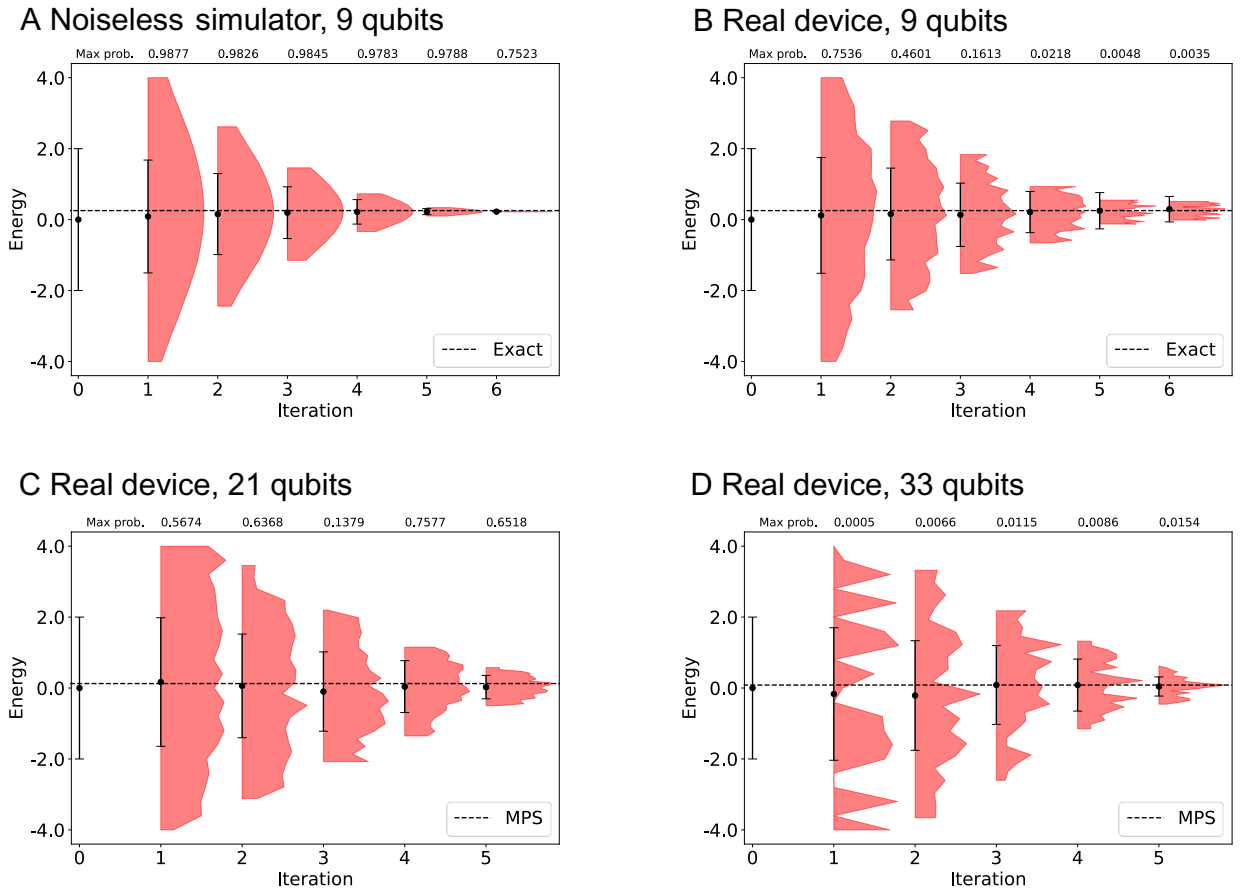


Fig. 3. Demonstration of our algorithm in the one-dimensional Hubbard models using (*A*) the noiseless simulator and (*B–D*) the Heron device *ibm_torino* with the error suppression. The circle and error bar denote the mean value and SD, respectively, of the posterior distribution in each iteration. The sampled probabilities are shown in red, and each plot is normalized by the maximum value in each iteration, which is shown at the *Top* of the plot.

previous section. As the iteration increases, the mean value of the posterior distribution (circle) approaches the exact energy (dashed line), and the SD (error bar) becomes smaller. The mean value of the posterior distribution in the final iteration is 0.224 ± 0.005, and the exact value is 0.254; i.e., the bias error 0.030 remains. From the results of $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ in the previous section, the accuracy is improved by increasing d_{evol} from five; in fact, if $d_{\mathrm{evol}} = 8$ and 10, the bias error decreases to 0.020 and 0.012, respectively. The probabilities taken from each circuit with ϵ determined from the prior distribution are shown in red, and a peak at the mean value can be seen.

Fig. 3 *B–D* show the results on the real device *ibm_torino*, where we used an error suppression module FireOpal in Q-CTRL (35–37). Certainly in Fig. 3*B*, the maximum probability of 0.75 at the first iteration was reduced to 0.0035 at the sixth iteration due to noise. Nevertheless, it was observed that the mean values move toward the exact solution, and the variance also decreases as the iterations increase. That is, our algorithm worked in the nine-qubit circuit, where the final value was 0.294 ± 0.359. The two-qubit gate count in the circuit in the final iteration is 2984/2274, where hereafter we regard the gate count value before and after slash as the value before and after the error suppression, respectively. Note that the error suppression was executed after the Qiskit transpilation.

Fig. 3 *C* and *D* are results for $n_{\mathrm{s}} = 10$ and 16 by using 21 and 33 qubit circuits, where $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle)$ in $d_{\mathrm{evol}} = 12$ are 0.92 and 0.81, respectively. d_{prep} was fixed to five since the values $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ tend not to depend on the model size, where the values are 4.6×10^{-3} in 20 system qubits, 4.5×10^{-3} in 32 qubits, and 5.3×10^{-3} even in 100 qubits. Although there are some numerical instabilities, such as in the low probability in the first iteration in Fig. 3*D*, we still obtain final gap values of 0.026 ± 0.330 and 0.043 ± 0.270 for 21 and 33 qubit circuits, and the MPS gap values for references are 0.125 and 0.084, respectively. As in *SI Appendix*, section S5, we theoretically confirm that our algorithm does not change the peak top value if a depolarization noise is assumed, which is the same as in the previous Bayesian QPE algorithms (6, 16). Furthermore, the effect of depolarization noise in a circuit execution exponentially reduces for N (with a fixed error rate). In fact, while the distribution for a nine-qubit model becomes blurred due to noise on the final iteration, sharp peaks can be found for 21- and 33-qubit models even though especially in the 33-qubit model, only 1.54% of the probability amplitude remains. Additionally, the number of two-qubit gates in the final iteration of the 33-qubit circuit is 7242/794, nearly 10 times gate reduction by the error suppression was found. As a result, our algorithm can be performed on more than five times larger systems in terms of the number of qubits than the previous QPE study of six qubit systems (14). Note that for the signal weakening in the first iteration on the 33 qubit demonstration, it is difficult for us to fully identify the cause due to the multiple error suppression components in FireOpal (*SI Appendix*, section S1), but it may be relevant that the number of gates in the circuit after the error

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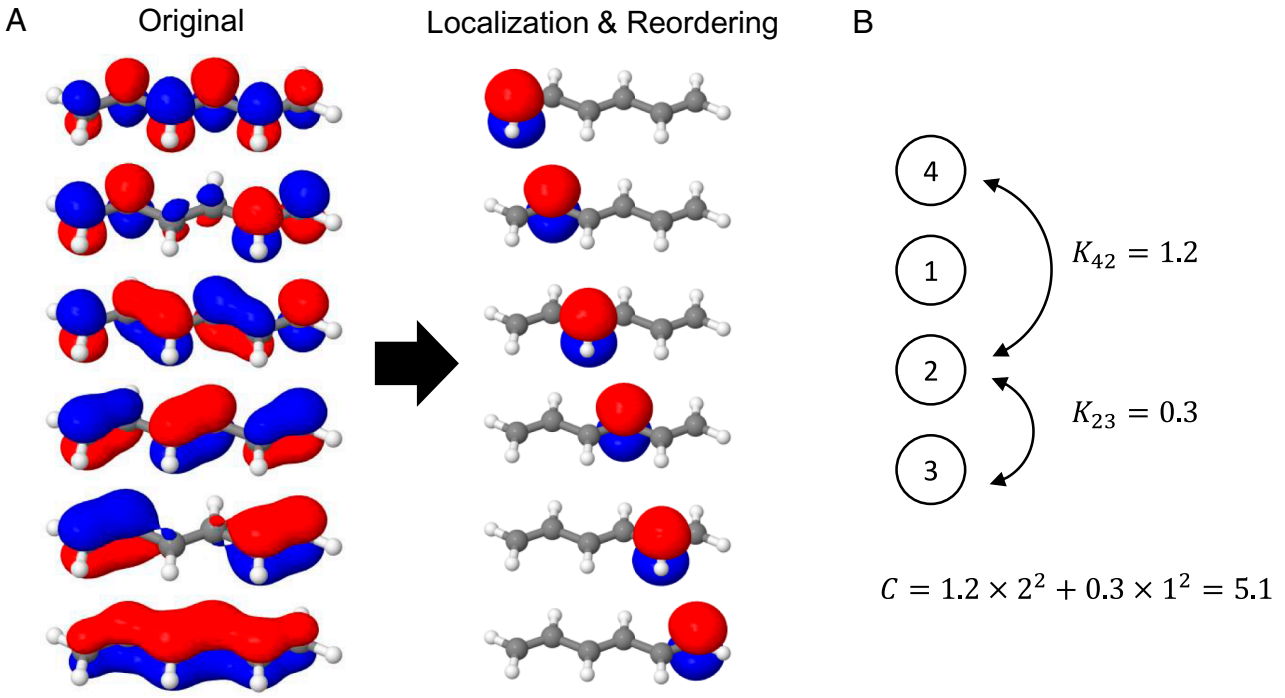


Fig. 4. Procedure for converting the molecular Hamiltonian to a one-dimensional one. (*A*) Orbitals before and after the procedure. The structures are drawn by Jmol (38). (*B*) Example of the cost function.

suppression was slightly high, where the gate counts in the first and final iterations were 819 and 794, respectively. We also note that there were only a few signals when the suppression module was not enabled, and the probabilities were too low to obtain meaningful results on the 21-qubit circuit (*SI Appendix*, section S5).

We next show the result of our algorithm to a realistic chemical model. The target molecule is hexatriene, and the Hamiltonian is a 12-qubit complete active space configuration interaction (6e 6o) problem consisting of π and π^* orbitals (*SI Appendix*, section S3). The obstacle to applying our algorithm in this model is that the molecular orbitals are delocalized throughout the molecule and thus the Hamiltonian is not one-dimensional. Therefore, we propose a general procedure to convert such Hamiltonian to a one-dimensional one, as shown in Fig. 4*A*. We first localized the orbitals by Boys localization (39). Then, the orbital indices are reordered by exchange interaction. Specifically, we introduce the cost function referencing the proposal in the context of tensor networks in classical calculations (40),

$$C = \sum_{ij} K_{ij} D_{ij}^2, \qquad [4]$$

where i and j are orbital indices, K_{ij} is the value of the exchange interaction between orbitals, and D_{ij} is the distance between orbitals when the orbital indices are arranged in a one-dimensional list. We show an example of C in Fig. 4*B*. C was optimized with a genetic algorithm (*SI Appendix*, section S3), and its initial and final values are 0.0754 to 0.0160, respectively. Finally, the reordered model was mapped to qubits by Jordan–Wigner transformation with the up–down qubit sequence.

We calculate the energy gap between the lowest spin-singlet and triplet states. The values of the metric $f(U_{\mathrm{prep}};\ |\mathrm{MPS}\rangle)$ in the original and in the one-dimensionalized models in $d_{\mathrm{prep}} = 10$ were 0.94 and 0.99, which indicates that U_{prep} is producing high fidelity despite the fact that both the ground and excited state cannot be approximated by a single configuration due to orbital localization. The values of $\delta(U_{\mathrm{evol}};\ U_{\mathrm{ref}})$ in $d_{\mathrm{evol}} = 8$ were 1.6×10^{-2} and 9.4×10^{-3}, respectively, more than 40% reduction in U_{evol}. Note that although the value of the improvement in the metric itself is not large, the improvement is significant because U_{evol} is applied $t/\Delta t$ times.

Fig. 5 shows the results of our algorithm for each model. The gap values in the original and one-dimensional models for the noiseless simulator in Fig. 5 *A* and *B* are -0.300 ± 0.008 and 0.236 ± 0.005, and the values for the real device in Fig. 5 *C* and *D* are -0.534 ± 1.112 and -0.185 ± 0.769, and the gate counts are 2190/1054 and 2718/658, respectively. The exact gap is 0.125 in this system. The results suggest that the one-dimensional processing can extend the range of applications of our algorithm. Note that we also executed octatetraene (decapentaene) of the localized orbital model on 17 (21) qubit circuits in *SI Appendix*, section S5. In octatetraene, we could not prepare U_{evol} of the molecular orbital model even calculating 300 h on a high-performance computer having more than 100 CPU cores and 1 TB RAM while that of the localized model can be calculated within two days using less than 20 CPU cores. This fact also suggests the improvements in our procedure. Furthermore, in order to compare the accuracy of the interaction complexity, we performed our algorithm on five structures of H_4 clusters that gradually changed from linear to square structures using noiseless simulations; see *SI Appendix*, section S7.

Conclusions and Outlook

We have proposed a gate-efficient and noise-robust quantum phase estimation-type algorithm based on unitary compression by tensor networks. We applied the algorithm to a one-dimensional Hubbard model and confirmed its effectiveness on a real superconducting device, utilizing an error suppression module, with

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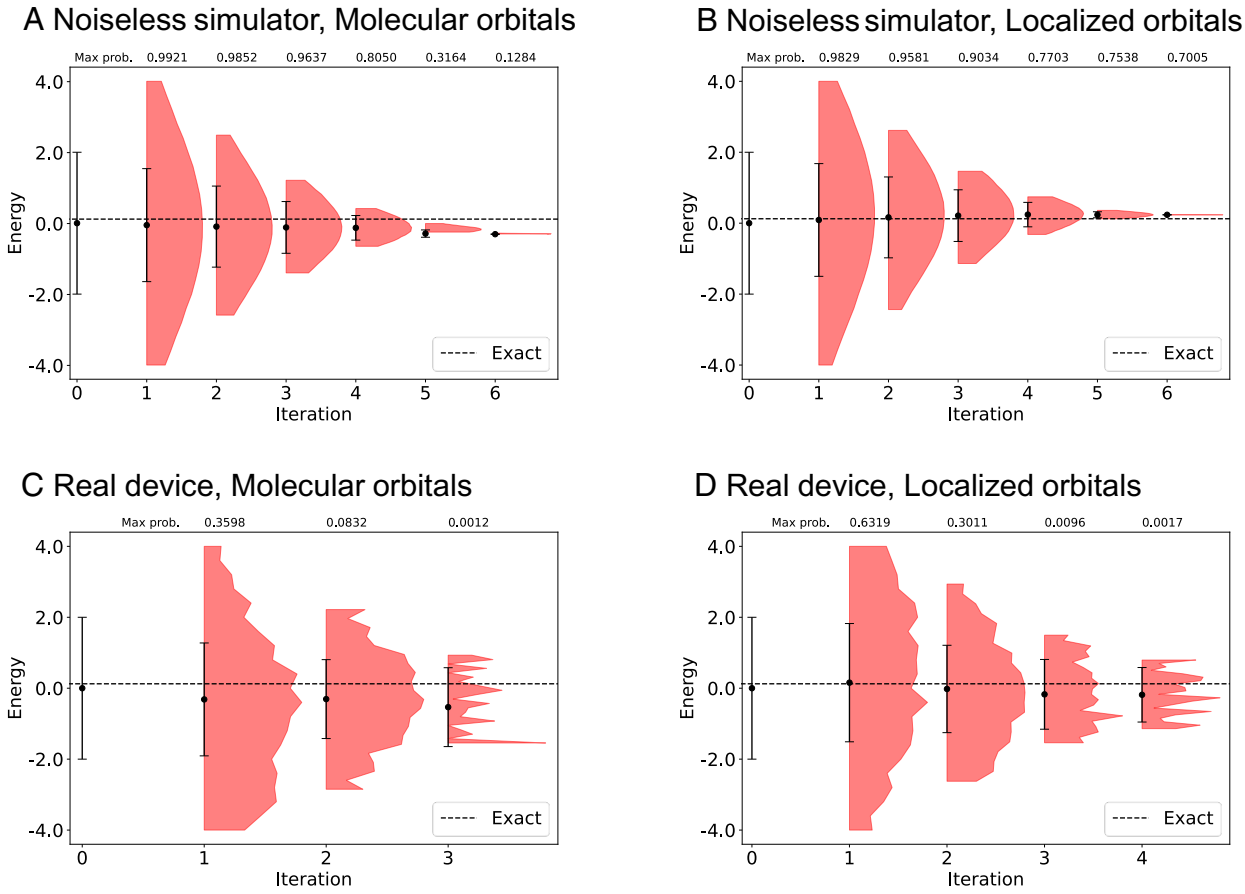


Fig. 5. Demonstration of our algorithm in hexatriene (13-qubit circuit) using (*A* and *B*) the noiseless simulator and (*C* and *D*) the Heron device *ibm_torino* with the error suppression. The circle and error bar denote the mean value and SD, respectively, of the posterior distribution in the iteration. The sampled probabilities are shown in red, and each plot is normalized by the maximum value in each iteration, which is shown at the *Top* of the plot.

circuits up to 33 qubits including one ancilla qubit. Furthermore, for the calculation of chemical models, we proposed a procedure to convert the model composed of molecular orbitals into a one-dimensional model by orbital localization and reordering, achieving over 40% improvement in the error metric. Based on these results, it has been demonstrated that QPE, which had previously been validated only in toy models with a small number of qubits, can be executed at a scale comparable to the classical computational limit of FCI, where the current limit is around 40 qubits, depending on the computing environment (41, 42).

The next step is to tackle tasks that are completely infeasible for classical computation, thereby achieving the so-called quantum advantage. The most critical challenge is an improvement of estimation to achieve chemical accuracy. A deeper MPO improves accuracy, but in general, the computational cost increases exponentially with depth. Since our unitary compression can be extended to tensor networks other than MPO, tensor networks that account for nonlinearity in device connectivity—for example heavy hex on IBM devices (43) or two-dimensional grids on Google devices (44). In the case of ion- or atom-trapped hardware, physical constraints on the transport of the particles would influence the optimal tensor structure for the algorithm. Alternatively, integrating quantum-classical hybrid-optimized circuit to U_{prep} can improve accuracy (22, 29, 45). Additionally, exploring the use of partially error-corrected devices anticipated in the near future could further advance the feasibility of our approach.

In the fully error-corrected era, the qubitization (46) becomes an alternative for approximating time evolution operator because it can realize the time evolution with $\mathcal{O}\left(\log\left(\epsilon^{-1}\right)\right)$ queries in contrast to the Trotter method of $\mathcal{O}\left(\mathrm{poly}\left(\epsilon^{-1}\right)\right)$ depth for accuracy ϵ while the qubitization is currently impractical due to a very high constant cost of the linear combination of unitaries (LCU). The classical circuit compression would also be useful even in the era, for example, in the PREPARE channel compression in LCU (47). It also should be mentioned that the Trotter method can achieve $\mathcal{O}\left(\mathrm{poly}\log\left(\epsilon^{-1}\right)\right)$ by using algorithmic error mitigations (48, 49).

Finally, in addition to QPE forming the basis of quantum computation, the technique we proposed for superposition state preparation to avoid long-range interactions involving an ancilla qubit can be used for more general circuit compilation (50). Hence, the proposed algorithms or techniques have potential applications in other fields besides chemistry, such as the Harrow–Hassidim–Lloyd algorithm (51) in machine learning.

Materials and Methods

The technical details of our algorithm are described in *SI Appendix*, sections S1–S3. *SI Appendix*, section S1 shows the procedure of our phase difference estimation algorithm and calculation conditions. *SI Appendix*, section S2 explains how to compress the initial state and time evolution operator via tensor networks. *SI Appendix*, section S3 describes the construction of the Hamiltonians in the Hubbard models and the linear polyenes.

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Data, Materials, and Software Availability. The codes and datasets in our study are available at: https://github.com/sk888ks/TQPDE_open.git (52). All other data are included in the manuscript and/or *SI Appendix*.

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Author affiliations: [a]Mitsubishi Chemical Corporation, Science & Innovation Center, Yokohama 227-8502, Japan; [b]Quantum Computing Center, Keio University, Yokohama 223-8522, Japan; [c]Graduate School of Science and Technology, Keio University, Kawasaki, Kanagawa 212-0032, Japan; [d]Centre for Quantum Engineering, Research and Education, TCG Centres for Research and Education in Science and Technology, Sector V, Salt Lake, Kolkata 700091, India; [e]Keio University Sustainable Quantum Artificial Intelligence Center, Keio University, Minato-ku, Tokyo 108-8345, Japan; [f]SoftBank Corp., Research Institute of Advanced Technology, Tokyo 105-7529, Japan; [g]Materials Informatics Initiative, RD Technology & Digital Transformation Center, JSR Corporation, Kawasaki 210-0821, Japan; and [h]Department of Applied Physics and Physico-Informatics, Keio University, Yokohama 223-8522, Japan

1. B. Bauer, S. Bravyi, M. Motta, G. Kin-Lic Chan, Quantum algorithms for quantum chemistry and quantum materials science. *Chem. Rev.* **120**, 12685–12717 (2020).
2. Y. Cao *et al*., Quantum chemistry in the age of quantum computing. *Chem. Rev.* **119**, 10856–10915 (2019).
3. A. J. Cohen, P. Mori-Sánchez, W. Yang, Challenges for density functional theory. *Chem. Rev.* **112**, 289–320 (2012).
4. A. Yu, Kitaev, Quantum measurements and the abelian stabilizer problem. arXiv [Preprint] (1995). https://doi.org/10.48550/arXiv.quant-ph/9511026 (Accessed 23 May 2025).
5. M. A. Nielsen, I. L. Chuang, *Quantum Computation and Quantum Information: 10th Anniversary Edition* (Cambridge University Press, Cambridge, England, 2010).
6. N. Wiebe, C. Granade, Efficient Bayesian phase estimation. *Phys. Rev. Lett.* **117**, 010503 (2016).
7. K. Yamamoto, S. Duffield, Y. Kikuchi, D. Muñoz Ramo, Demonstrating Bayesian quantum phase estimation with quantum error detection. *Phys. Rev. Res.* **6**, 013221 (2024).
8. B. P. Lanyon *et al*., Towards quantum chemistry on a quantum computer. *Nat. Chem.* **2**, 106–111 (2010).
9. J. Du *et al*., NMR implementation of a molecular hydrogen quantum simulation with adiabatic state preparation. *Phys. Rev. Lett.* **104**, 030502 (2010).
10. Y. Wang *et al*., Quantum simulation of helium hydride cation in a solid-state spin register. *ACS Nano* **9**, 7769–7774 (2015).
11. P. J. J. O'Malley *et al*., Scalable quantum simulation of molecular energies. *Phys. Rev. X* **6**, 031007 (2016).
12. S. Paesani *et al*., Experimental Bayesian quantum phase estimation on a silicon photonic chip. *Phys. Rev. Lett.* **118**, 100503 (2017).
13. R. Santagati *et al*., Witnessing eigenstates for quantum simulation of Hamiltonian spectra. *Sci. Adv.* **4**, eaap9646 (2018).
14. N. S. Blunt, L. Caune, R. Izsák, E. T. Campbell, N. Holzmann, Statistical phase estimation and error mitigation on a superconducting quantum processor. *PRX Quantum* **4**, 040341 (2023).
15. K. Sugisaki *et al*., Bayesian phase difference estimation: A general quantum algorithm for the direct calculation of energy gaps. *Phys. Chem. Chem. Phys.* **23**, 20152–20162 (2021).
16. K. Sugisaki *et al*., Quantum algorithm for full configuration interaction calculations without controlled time evolutions. *J. Phys. Chem. Lett.* **12**, 11085–11089 (2021).
17. Y. Matsuzaki, H. Hakoshima, K. Sugisaki, Y. Seki, S. Kawabata, Direct estimation of the energy gap between the ground state and excited state with quantum annealing. arXiv [Preprint] (2020). https://doi.org/10.48550/arXiv.2007.10561 (Accessed 23 May 2025).
18. A. E. Russo, K. M. Rudinger, B. C. A. Morrison, A. D. Baczewski, Evaluating energy differences on a quantum computer with robust phase estimation. *Phys. Rev. Lett.* **126**, 210501 (2021).
19. M. S. Rudolph, J. Chen, J. Miller, A. Acharya, A. Perdomo-Ortiz, Decomposition of matrix product states into shallow quantum circuits. *Quantum Sci. Technol.* **9**, 015012 (2023).
20. S. J. Ran, Encoding of matrix product states into quantum circuits of one- and two-qubit gates. *Phys. Rev. A* **101**, 032310 (2020).
21. S. H. Lin, R. Dilip, A. G. Green, A. Smith, F. Pollmann, Real- and imaginary-time evolution with compressed quantum circuits. *PRX Quantum* **2**, 010342 (2021).
22. J. Dborin, F. Barratt, V. Wimalaweera, L. Wright, A. G. Green, Matrix product state pre-training for quantum machine learning. *Quantum Sci. Technol.* **7**, 035014 (2022).
23. J. Dborin *et al*., Simulating groundstate and dynamical quantum phase transitions on a superconducting quantum computer. *Nat. Commun.* **13**, 5977 (2022).
24. C. Mc Keever, M. Lubasch, Classically optimized Hamiltonian simulation. *Phys. Rev. Res.* **5**, 023146 (2023).
25. B. Anselme Martin, T. Ayral, F. Jamet, M. J. Rančić, P. Simon, Combining matrix product states and noisy quantum computers for quantum simulation. *Phys. Rev. A* **109**, 062437 (2024).
26. L. Causer, F. Jung, A. Mitra, F. Pollmann, A. Gammon-Smith, Scalable simulation of nonequilibrium quantum dynamics via classically optimized unitary circuits. *Phys. Rev. Res.* **6**, 033062 (2024).
27. T. Shirakawa, H. Ueda, S. Yunoki, Automatic quantum circuit encoding of a given arbitrary quantum state. *Phys. Rev. Res.* **6**, 043008 (2024).
28. C. Mc Keever, M. Lubasch, Towards adiabatic quantum computing using compressed quantum circuits. *PRX Quantum* **5**, 020362 (2024).
29. A. Peruzzo *et al*., A variational eigenvalue solver on a photonic quantum processor. *Nat. Commun.* **5**, 4213 (2014).
30. K. Nakayama, H. Nakano, K. Hirao, Theoretical study of the $\pi \rightarrow \pi^*$ excited states of linear polyenes: The energy gap between 1^1B_u+ and 2^1A_g- states and their character *Int. J. Quantum Chem.* **66**, 157–175 (1998).
31. K. R. Brown, J. Kim, C. Monroe, Co-designing a scalable quantum computer with trapped atomic ions. *NPJ Quantum Inf.* **2**, 1–10 (2016).
32. S. R. White, Density matrix formulation for quantum renormalization groups. *Phys. Rev. Lett.* **69**, 2863–2866 (1992).
33. U. Schollwöck, The density-matrix renormalization group in the age of matrix product states. *Ann. Phys.* **326**, 96–192 (2011).
34. A. Javadi-Abhari *et al*., Quantum computing with Qiskit. arXiv [Preprint] (2024). https://doi.org/10.48550/arXiv.2405.08810 (Accessed 23 May 2025).
35. P. S. Mundada *et al*., Experimental benchmarking of an automated deterministic error-suppression workflow for quantum algorithms. *Phys. Rev. Appl.* **20**, 024034 (2023).
36. N. Sachdeva *et al*., Quantum optimization using a 127-qubit gate-model IBM quantum computer can outperform quantum annealers for nontrivial binary optimization problems. arXiv [Preprint] (2024). https://doi.org/10.48550/arXiv.2406.01743 (Accessed 23 May 2025).
37. Y. Baum *et al*., Experimental deep reinforcement learning for error-robust gate-set design on a superconducting quantum computer. *PRX Quantum* **2**, 040324 (2021).
38. Jmol Development Team, Jmol: an open-source Java viewer for chemical structures in 3D. https://jmol.sourceforge.net/. Accessed 23 May 2025
39. J. M. Foster, S. F. Boys, Canonical configurational interaction procedure. *Rev. Mod. Phys.* **32**, 300–302 (1960).
40. N. Nakatani, G. K. L. Chan, Efficient tree tensor network states (TTNS) for quantum chemistry: Generalizations of the density matrix renormalization group algorithm. *J. Chem. Phys.* **138**, 134113 (2013).
41. Q. Sun *et al*., Recent developments in the PySCF program package. *J. Chem. Phys.* **153**, 024109 (2020).
42. H. Gao, S. Imamura, A. Kasagi, E. Yoshida, Distributed implementation of full configuration interaction for one trillion determinants. *J. Chem. Theory Comput.* **20**, 1185–1192 (2024).
43. Y. Kim *et al*., Evidence for the utility of quantum computing before fault tolerance. *Nature* **618**, 500–505 (2023).
44. F. Arute *et al*., Quantum supremacy using a programmable superconducting processor. *Nature* **574**, 505–510 (2019).
45. K. Nakaji *et al*., The generative quantum eigensolver (GQE) and its application for ground state search. arXiv [Preprint] (2024). https://doi.org/10.48550/arXiv.2401.09253 (Accessed 23 May 2025).
46. G. H. Low, I. L. Chuang, Hamiltonian simulation by qubitization. *Quantum* **3**, 163 (2019).
47. H. Morisaki, K. Mitarai, K. Fujii, Y. O. Nakagawa, Classical variational optimization of a PREPARE circuit for quantum phase estimation of quantum chemistry Hamiltonians. *Phys. Rev. Res.* **6**, 043186 (2024).
48. S. Endo, Q. Zhao, Y. Li, S. Benjamin, X. Yuan, Mitigating algorithmic errors in a Hamiltonian simulation. *Phys. Rev. A* **99**, 012334 (2019).
49. J. D. Watson, J. Watkins, Exponentially reduced circuit depths using trotter error mitigation. arXiv [Preprint] (2024). https://doi.org/10.48550/arXiv.2408.14385 (Accessed 23 May 2025).
50. S. Kanno *et al*., Quantum computing quantum Monte Carlo with hybrid tensor network for electronic structure calculations. *NPJ Quantum Inf.* **10**, 1–13 (2024).
51. A. W. Harrow, A. Hassidim, S. Lloyd, Quantum algorithm for linear systems of equations. *Phys. Rev. Lett.* **103**, 150502 (2009).
52. S. Kanno, "TQPDE_open." GitHub, https://github.com/sk888ks/TQPDE_open.git. Accessed 22 May 2025.

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